1 Analysis and Classification of a Biomechanical Signal

This activity regards signals analysis and their classification based on a supervised learning method which let recognize the type of considered signal. The involved process starts from the signal acquisition and it's carried on by the processing of data with the software LabView and MatLab.

1.1 Aim of the exercise

Getting into the details of the exercise, the focus is the analysis of different acceleration signals in time and frequency domains for three different activities: standing, walking and running.
The primary goal is the detection of the main features of these signals to understand the analogies and differences between them, by computing statistical parameteres that are considered significant for this comparison. Subsequently, using data collected from other users, a supervised learning method must be implemented in order to classify one or more *Test* signals into one of the three activities, based on the features extracted from the *Training* data.

1.2 Materials and methods

The followed procedures to accomplish the different tasks are described in this section. In the first part the attention is given to the acquisition and analysis of the acceleration signals and their description. The second part considers the whole set of data and the implementation of the supervised learning method for classification.

1.2.1 Acquisition & Pre-processing

The acceleration data are acquired with the app Phyphox which can record the signal for a fixed time by extracting the information of the accelerometer sensors placed in the mobile phone which returns the values in the three directions (x,y,z), defined by default as visible in Figure 1. The gravity acceleration is also taken into account, although the app offers the possibility to neglect it.

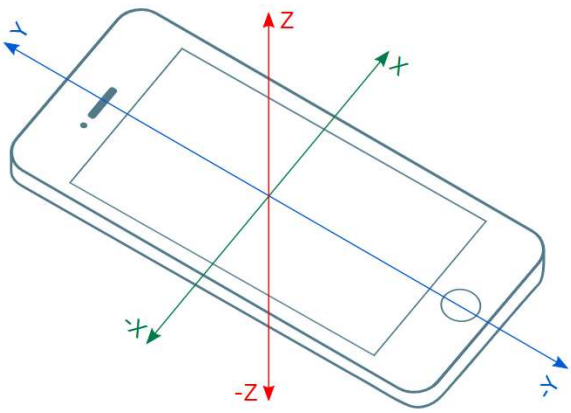


Figure 1. Mobile phone reference frame (*Fons*)

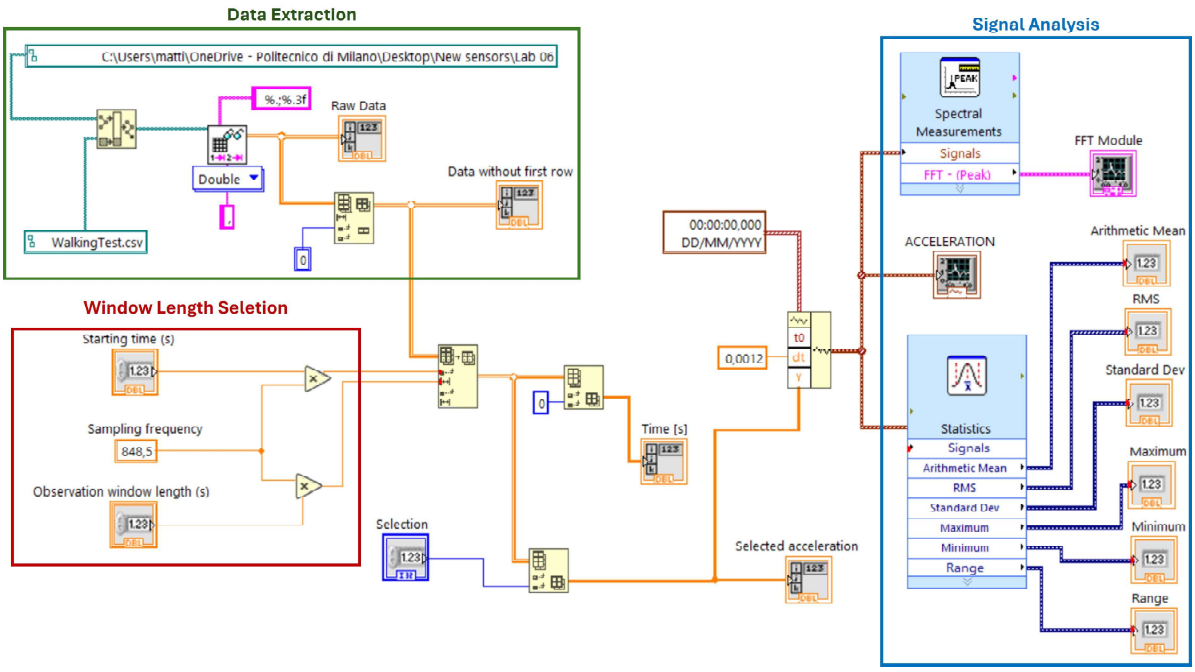


Figure 2. Block Diagram Labview

The provided output of Phyphox is a *.csv* file which collects by columns the time instants and the corresponding values of the acceleration in each direction and its absolute value in the last column. This file is the input of the developed LabView code (Figure 2). For istance, the selected activity is a normal walking characterized by a duration of 15 seconds.
The 'Data extraction' part of the code has the intention to extract the data through few steps:

- Provide the path and the corresponding file name of the recorded accelerations.
- Specify the delimiter for the columns (*comma*,',') and the significant digits (3).
- Remove the first row that is dedicated to the column headers.

The outcome of this process is a matrix $(n \cdot 5)$ containing 5 columns collecting respectively the time, the x,y,z accelerations and the absolute value for each of the n-time steps. Subsequently, the duration of the signal and the acceleration component are selected by the user which imposes the starting time, the window length and the column corresponding to the desired acceleration component (1,2,3) in the 'Window Length Selection' part.
In order to make this selection the sampling frequency has to be detected since the selected number of samples from the matrix depends on f_s. Once the time vector is extracted the difference between each time step is computed and the mean value represents the Δt, which is the inverse of the sampling frequency. In this specific case, the measurements are taken with a sampling frequency $f_s = 848.5$ Hz.
Finally, the set time vector and the corresponding selected acceleration component are plotted with the *Build Waveform* block in the right part of the interface, followed by the 'Signal Analysis' window. For the time domain, the *Statistics* block is added to compute some features related to the signal (Mean, standard deviation, RMS, maximum,

minimum and range) and concerning the frequency domain, the FFT is computed with a basic rectangular window through the *Spectral Measurements* block.
These operations are repetead for three different activities which are standing, walking and running, repeated three times each, resulting in 9 different acquisitions. The reason of taking into account three repetitions of the same movement lies into the fact that the data's features are more representative of the considered activity.

1.2.2 Data Classification

The acquisition procedure of the previous section is made by other 15 different subjects, defining the whole set of data considered for the classification exercise. Once all the signals are stored, the processing of the data and their classification is developed on Matlab. After collecting all the signals, time and frequency features are extracted in order to implement a supervised learning method. The extracted features are:

1. Mean: this value is taken in absolute value in order to avoid the sign mismatch between the subjects due to the positioning of the phone in the pocket.
2. Root mean square (RMS);
3. Standard deviation (SD);
4. Maximum value;
5. Minimum value;
6. Range (peak-to-peak amplitude);
7. First frequency peak ("fundamental frequency").

Each of these 7 features is computed for the three directions, resulting in a total number of 21 features. The entire dataset is composed by a number $N = P \cdot R \cdot 3 = 144$ where P corresponds to the number of subjects (16), R the number of repeated measurements (3). Therefore, the overall feature matrix obtained is made by a $144 \cdot 23$ elements: the number of columns is composed by the 21 features, plus the subject ID and the activity class (standing, walking, running) indicated by a number from 1 to 3. For what concerns the time features extraction, there are already implemented Matlab functions that compute the statistic parameters; instead the first frequency peak is detected as reported in the section A.1. For this purpose, the peaks are researched with the function *findpeaks* by imposing a magnitude that is the 25% of the maximum peak, excluding the zero term frequency which affects in most of the cases (expecially in the gravity direction) the detection of the first peak which has not a sufficient amplitude to be included in the treshold.
Once the feature matrix is composed, a KNN (K-nearest-neighbors) algorithm is implemented in order to classify a selected number of test data that take part to the entire dataset. A further investigation is made in order to assess how the number of training and test data affects the accuracy of the classification.

1.3 Results and discussion

The obtained results of this exercise are reported and discussed in two separated parts, considering the analysis of acceleration signals for each activity in the first part and the classification of dataset based on the activity class in the second part.

1.3.1 Signal Analysis

The activities involved in the study are plotted in time and frequency domains and the corresponding statistic parameters are computed for each time domain signals. Since for each activity the acceleration recording is repeated three times, the mean, the root mean square (RMS), the standard deviation, the maximum, the minimum and the range are computed as a mean of the values obtained for each repetition of the same signal. Regarding each activity, the component (x,y,z) that is detected as the most significant is reported in the graphs in order to not burden the description.

Standing: Starting with the simplest case, it can be assessed that the oscillation visible in the standing case is related to the posture maintenance of the subject. Cause the small difference between the acceleration components, as an example the Figure 3 is reported and it describes the z component behavior in time and frequency domains.

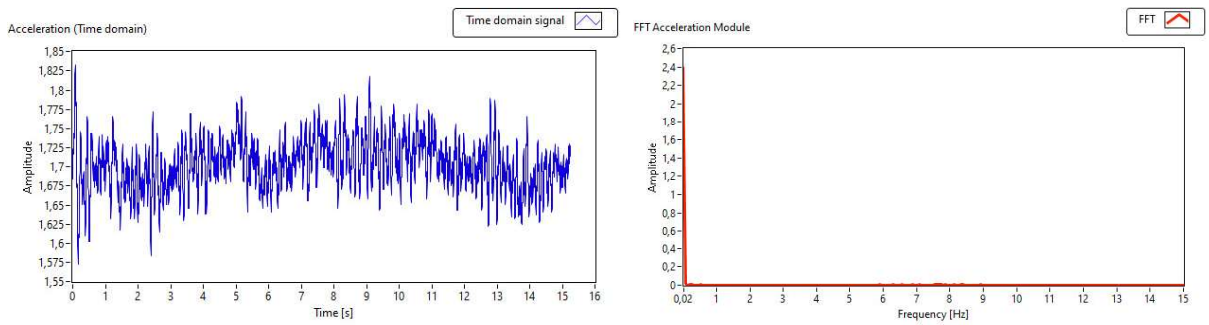


Figure 3. 1st Standing acceleration signal - Z component

By considering the trend of the signal in time, it's visible that the acceleration is featured by oscillations that develope around the mean of the signal (around 1.7 $\frac{m}{s^2}$) and they are small in terms of variation since the posture is easily maintained by the subject.
In the Table 1 all the statistic parameters of this activity are reported. Although the mean is very different (for instance the gravity contribution for y axis) the standard deviation and the range are very small, since the subject is only maintaining its posture.

STANDING	Mean	RMS	SD	Max	Min	Range
Acceleration X $[\frac{m}{s^2}]$	0.62	2.03	0.03	0.75	0.50	0.24
Acceleration Y $[\frac{m}{s^2}]$	-9.44	9.44	0.02	-9.35	-9.53	0.18
Acceleration Z $[\frac{m}{s^2}]$	1.13	1.13	0.05	1.33	0.92	0.41

Table 1. Statistic parameters for Standing activity

From a frequency point of view, all the standing acceleration signals present a zero-frequency component with an amplitude which is approximately the mean of the signal in time domain and there are no interesting frequency components in the spectrum due to the their very small amplitude.

Walking: Passing to the second activity, the considerations become a bit different since the signal involves a dynamic movement. In this case, the x acceleration component is reported in the Figure 4.

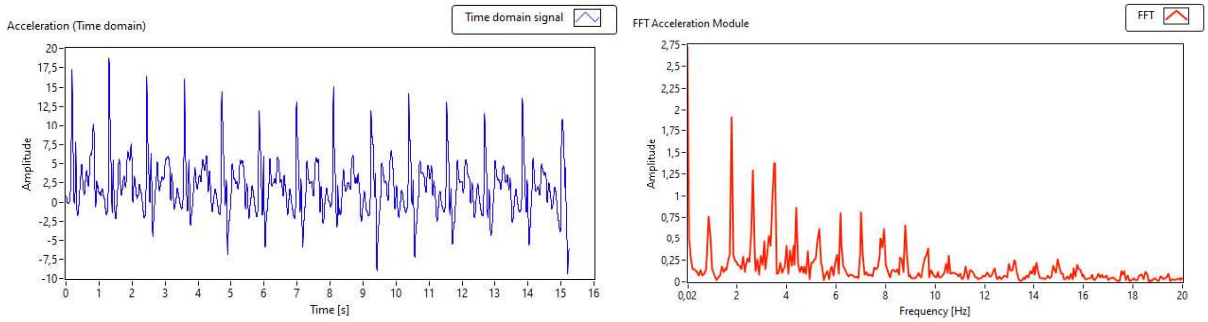


Figure 4. 1st Walking acceleration signal - X component

From a first inspection, the signal in time domain seems to be almost periodic since a well visible peak is repeated around every 1.2 seconds and this aspect can be confirmed in the frequency spectrum in which there is a peak in correspondence of about 0.8 Hz, which is the inverse of the delta in time. Since the mobile phone is placed in a pocket, it can be hypothesized that these peaks represent the moments at which the heel enters in contact with the ground (stride frequency), causing a huge difference in terms of acceleration. Moreover, the highest peak in the frequency domain is detectable at about 1.6 Hz (corresponding to 0.6 seconds of delta in time) which could be the step frequency, which it's a reasonable value for a normal velocity walking. In general the spectrum presents peaks at frequencies that are almost integer multiple of $f = 0.8$ Hz but this value is not properly the fundamental frequency since the signal is not periodic. That frequency can be defined as an *apparent fundamental frequency*, since the signal can be defined quasi-periodic. Similar considerations can be made for the other two acceleration components that present the peaks at the same distance (in time and frequency).

WALKING	**Mean**	**RMS**	**SD**	**Max**	**Min**	**Range**
Acceleration X $[\frac{m}{s^2}]$	1.9	4.4	3.3	16	-9.4	25.4
Acceleration Y $[\frac{m}{s^2}]$	-9.7	10.9	5.1	1.5	-32.1	33.7
Acceleration Z $[\frac{m}{s^2}]$	-0.79	7.6	7.5	12.6	-60.8	73.4

Table 2. Statistic parameters for Walking activity

Finally, the Table 2 shows the main statistic parameters of the walking activity. As visible, the mean is similar to the standing case for each component but obviously the range is much larger, expecially for the z component which is the one interested by

relevant movements during the swing phase due to the positioning of the sensor (mobile phone).

Running: The final activity is more similar to the second one since the acceleration measurements regard dynamic recordings. Following the same statements done for the walking activity, it's visible that the peaks in the frequency domain are bigger in correspondence of almost integer multiple of $f = 1.4$ Hz which seems to be the apparent fundamental frequency of the running activity. It's reasonable that this value is higher than the one detected for the walking activity (almost 2 times bigger) due to the fact that, in the same amount of time, the subject takes more steps, resulting in a higher stride frequency.

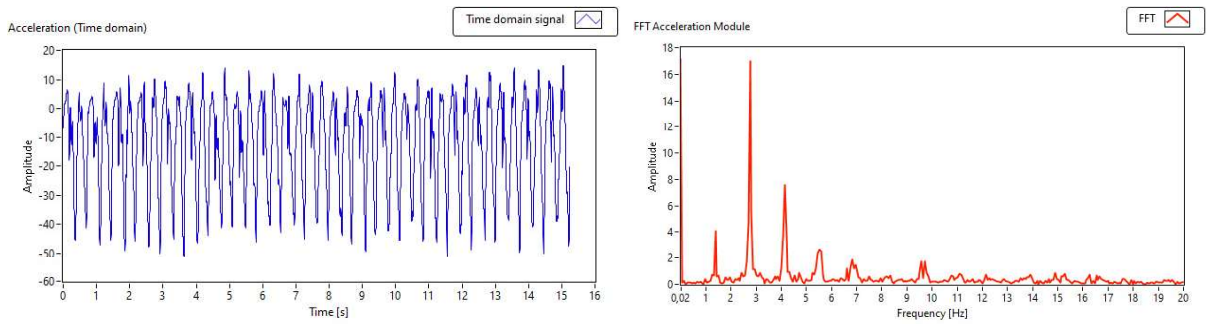


Figure 5. 2nd Running acceleration signal - Y component

For this purpose, the y acceleration component is shown in the Figure 5 in which the frequency peaks can be easily found, included the one at zero-frequency, featured by an amplitude related to the mean value.

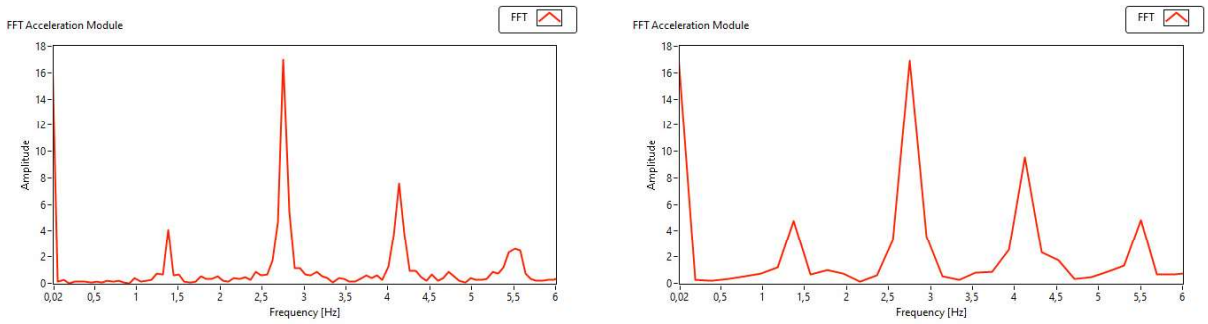
RUNNING	Mean	RMS	SD	Max	Min	Range
Acceleration X $[\frac{m}{s^2}]$	2.1	10.5	10.3	29.6	-22.5	52.0
Acceleration Y $[\frac{m}{s^2}]$	-12.2	20.9	17.0	26.6	-58.0	84.6
Acceleration Z $[\frac{m}{s^2}]$	0.0	18.0	18.0	33.7	-78.2	111.9

Table 3. Statistic parameters for Running activity

To conclude this paragraph, the Table 3 shows the main statistic parameters of the acceleration signals which are featured by a huge range of oscillation due to the magnitude of inertial and contact forces that act in the running activity.

Final considerations: From an analytical point of view, the acceleration signals are not perfectly periodic. Therefore the fundamental frequency is equal to the inverse of the window length: in this case $f_0 = \frac{1}{15s} = 0.067$ Hz, since the window length is set to 15 seconds. This fundamental frequency is also the frequency resolution of the spectra; for this reason, by changing the zoom (in time) the resolution changes and becomes smaller for reduced window lengths. These statements can be confirmed by the Figure 6 in which two different windows are applied to the signal, changing the frequency resolution; in

fact, the peaks are still present but with a wider bandwidth. In the case of applying a much smaller window length, the leakage phenomenon could be observed.



(a) Larger window length: $T = 15s$

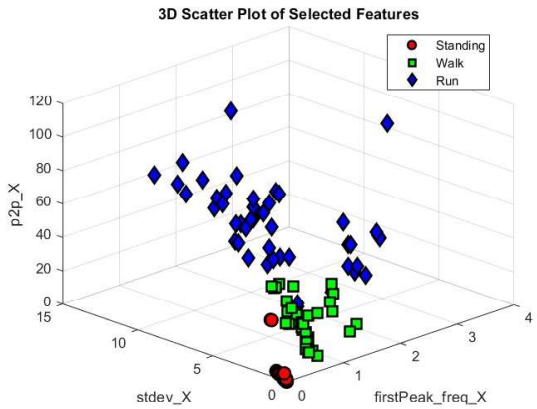
(b) smaller window length: $T = 5s$

Figure 6. 2nd Running acceleration signal - Y component

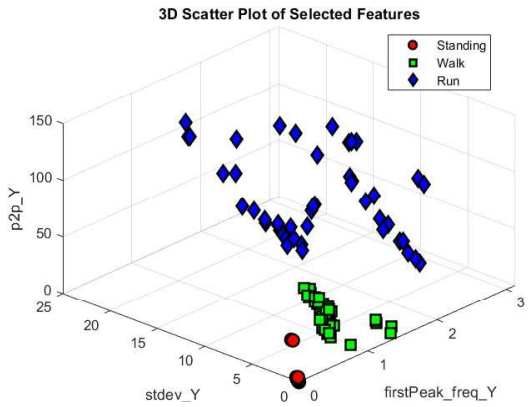
1.3.2 Data Classification

Once the feature matrix is built, some critical analyses can be made. Although the computation of different features give an expanded view of all the data, there are some statistic parameters which are strictly dependent on the subjects and on the way they performed the data acquisition. For instance, the maximum and minimum of the time signals can vary a lot between the subjects and it can be influenced by the mobile phone positioning, making the range of acceleration more meaningful for the classification. Another important parameter could be the first frequency peak which (if correctly detected) can discriminate very well the class of the activity. Moreover, the mean could not be so determinant to classifiy the data due to the small variation of this value; opposite obeservations can be made for the standard deviation which is more sensitive. Starting from these considerations, a visual check can be made by computing 3D plots which consider three different features to see how the data are distributed.
With the aim to show three different features not directly related with each other and not strictly dependent to the considered acceleration component, in the Figure 7 there are scatter plots that collect the data based on the standard deviation, the range and the first frequency peak (Figure 7a, 7b, 7c), which are reasonable features. This sentence can be proved by seeing that the points divided per activity tend to distribute in different zones depending on the class. On the other hand in the last plot (7d) there is an example of features which are not very useful tools to classify a dataset: the mean is not a reliable feature in this context.
To conclude the study on data classification, the study focuses on the effect of testing data amount and the choice of the K-value. In order to complete this task, some possibilities are studied:

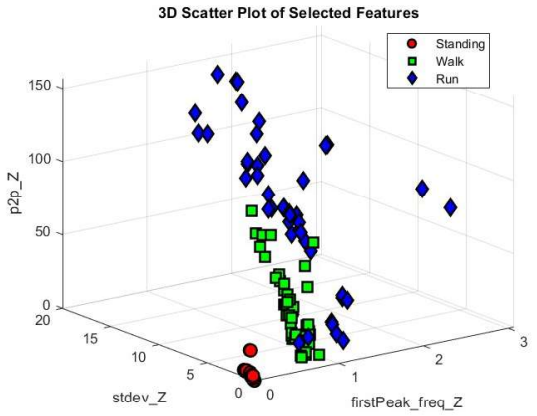
- The number of testing data are considered in 16 blocks (corresponding to the number of subjects) and the range considered goes from one subject to 14 subjects to test;
- The values of K consider a series of odd numbers from 1 to 29, resulting in 15 possibilities.



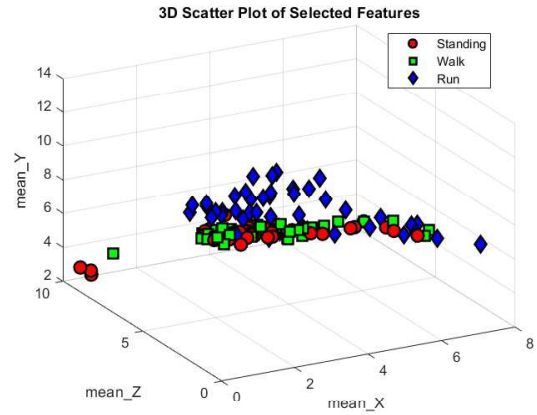
(a) SD, Frequency peak, Range - x component



(b) SD, Frequency peak, Range - y component



(c) SD, Frequency peak, Range - z component



(d) Mean x, mean y, mean z

Figure 7. Scatter plot of the features

To assess the performance of the algorithm the accuracy is computed by considering the total number of correct classifications divided by the amount of tested data. In the Figure 8 is shown the distribution of the accuracy depending on these two variables. For a better visualization, the accuracy values higher than 85% are visible in green, instead those lower than 50% in red.

In order to understand the contributions of these input parameters, additional graphs are reported. In the Figure 9a all the data are plotted and it's possible to see the trend of the accuracy depending on the number of testing data, which is complementary to the number of training data. As expected, with a large amount of testing samples the algorithm does not perform very well, since the training data are too low to guarantee a solid classification. Also with few testing samples the algorithm is not so efficient and the accuracy can vary in a wide range of values. The results start to be enclosed in smaller ranges when the number of test data increases. The best accuracy is found when this number goes between 60 and 100, corresponding to test from 7 subjects to 11 subjects. It seems that using approximately the 50% of the data to train the model, the algorithm has sufficient data to classify the remaining signals. For what concerns the Figure 9b, it's possible to assess that, considering the total amount of data (144) a value

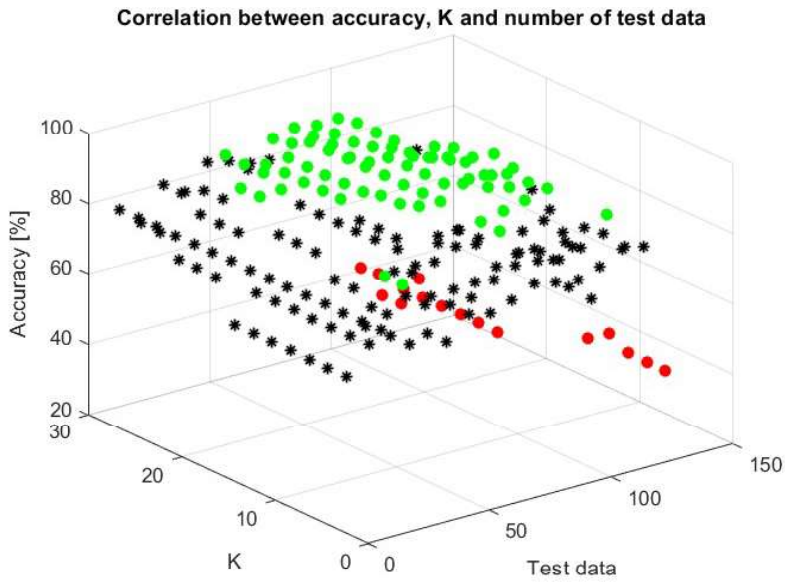
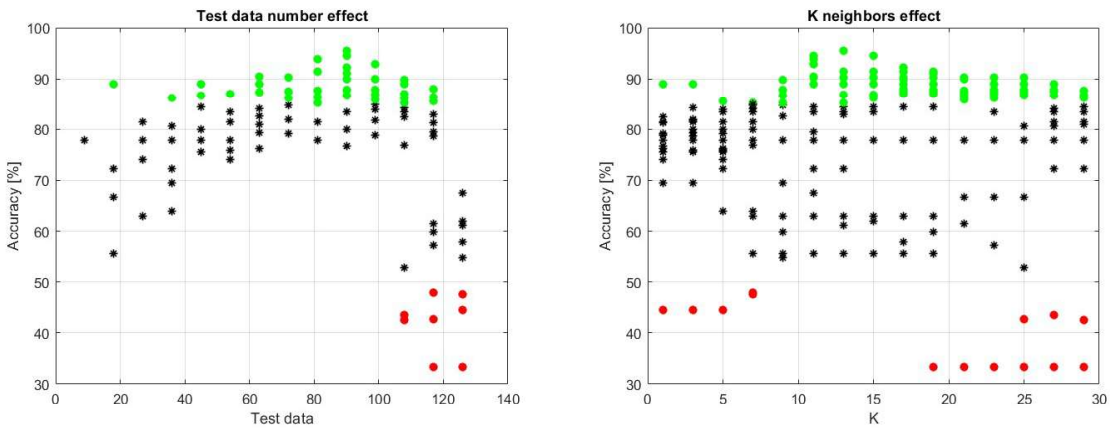


Figure 8. Accuracy distribution

of K around 15 is a good trade-off to have a sufficient classification, even if K is also related to the number of testing data. Moreover, in this case low or high values of K don't perform in a proper manner.
Although these results, the evaluation of the performance through the accuracy computation is limited and more features could be used to assess the overall efficiency. Additionally, the algorithm is strictly dependent on the quality of the provided data and in this case the order of the signals collected in the dataset was not changed, leading to a higher dependency from the first signals. Further investigations could be made to fine tune the code and give as input to the algorithm a more representative feature matrix.



(a) Testing data effect

(b) K value effect

Figure 9. KNN algorithm